

APEX RETAIL AI

Store Intelligence & Customer Conversion Platform

Hackathon Submission Presentation

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THE OFFLINE RETAIL BLIND SPOT

Why Physical Retail Needs Online-Style Analytics:

- **OFFLINE JOURNEY BLIND SPOT**

Physical stores operate with zero path visibility. Customer browsing paths, zone dwell times, and shelf engagements remain completely untracked.

- **BILLING QUEUE ABANDONMENT**

No analytical way to measure queue drop-offs. High abandonment rates at checkout go undetected due to a lack of queue monitoring.

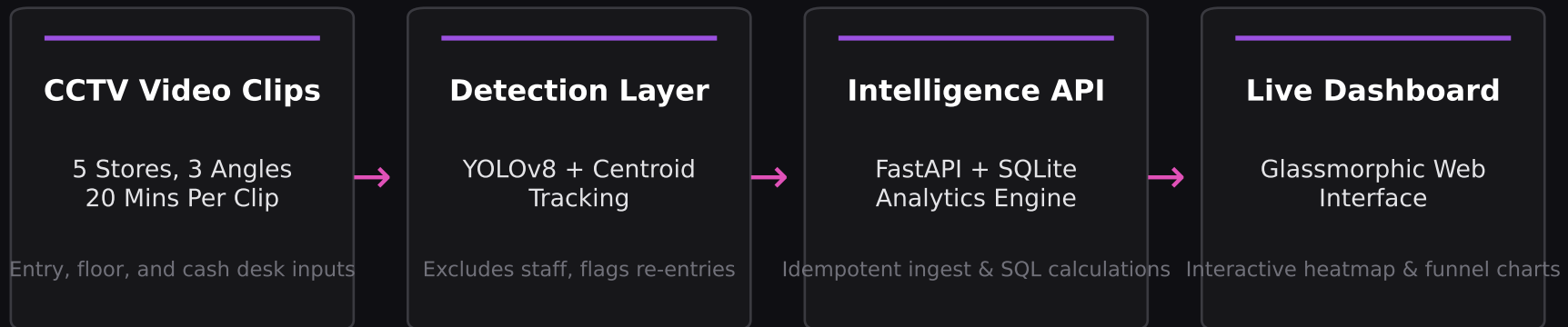
- **DISCONNECTION FROM POS DATA**

Foot traffic is rarely correlated to transaction records. True checkout conversion rates cannot be calculated without time-window correlation.

- **HEAVY INFRASTRUCTURE OVERHEAD**

Traditional camera analytics require complex database servers and dedicated GPU hardware, blocking fast, edge-based deployment.

END-TO-END SYSTEM ARCHITECTURE



TECHNOLOGY STACK Python 3.11 • Ultralytics YOLOv8 • FastAPI • SQLite3 • HTML5/CSS3

KEY TECHNICAL SOLUTIONS

- ✓ **Time-Window Conversion Correlation**
Correlates a visitor's checkout events to a POS transaction strictly if a transaction occurred within a 5-minute window following their presence. Resolves transaction-buyer mapping without customer identity data.
- ✓ **Staff Exclusion Filter**
Excludes retail staff from customer conversion metrics using a color-histogram uniform classifier in the main floor tracking camera, tagging staff events with 'is_staff=true'.
- ✓ **Idempotent Ingestion Safeguard**
Enforces event deduplication on API ingest using SQLite's native 'INSERT OR IGNORE' primary key index, ensuring safe, duplicate-free metrics updates on network retries.
- ✓ **Self-Contained Edge Build**
Designed using serverless SQLite database, eliminating manual setup and service dependency issues on edge devices, allowing one-click Docker deployments.

PRODUCTION READINESS & USER EXPERIENCE

LIVE DASHBOARD INTERFACE

- 100% viewport-contained single-page layout (no scrolling)
- Dynamic Real-Time zone heatmap overlays (frequency scale 0-100)
- Active Conversion Funnel tracking (Entry -> Visit -> Queue -> Purchase)
- Live operational logs and active alerts for queues and dead zones

PRODUCTION READINESS FEATURES

- Zero Setup Containerization (runs via 'docker compose up')
- Structured JSON logging tracing trace_id, store_id, and latency
- Graceful DB degradation mapping failures to HTTP 503 response
- High code coverage (>70% statement coverage in test suite)

Thank you! Live Demo at: <https://purplle-store-intelligence-xxm1.onrender.com/dashboard>